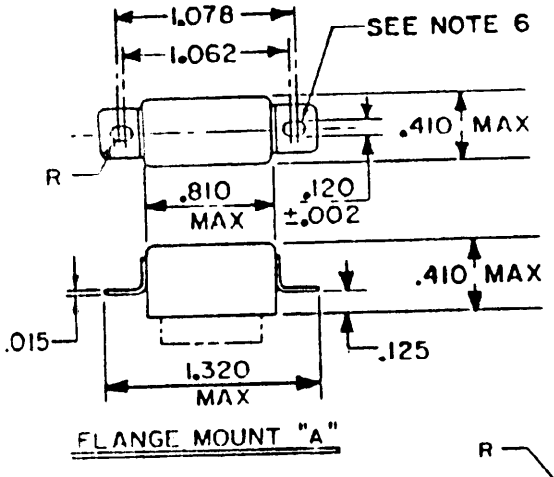
MILITARY SPECIFICATION SHEET

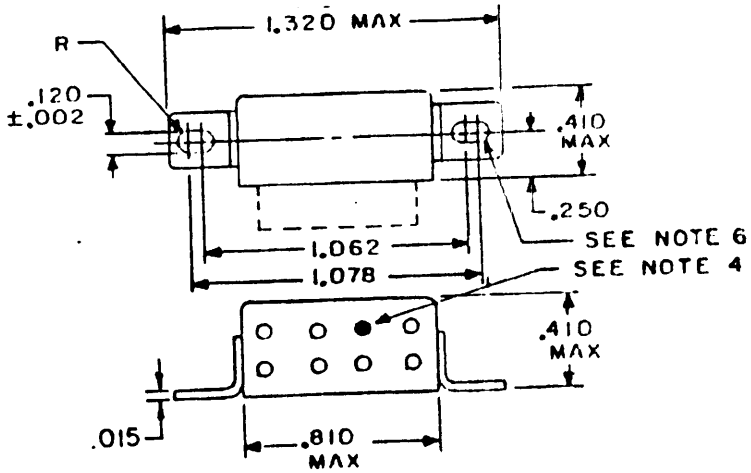
RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2 AMPERES (0.200-INCH TERMINAL SPACING), INCLUDING INSPECTIONS OF APPENDIXES A AND B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016.



FLANGE MOUNT "A"



FLANGE MOUNT "B"

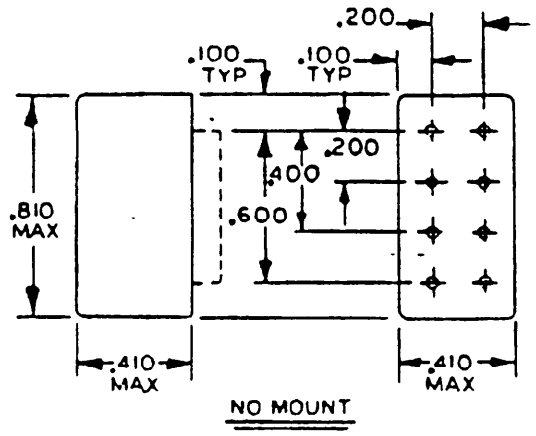
FIGURE 1. Dimensions and configurations.

FSC 5945

AMSC N/A

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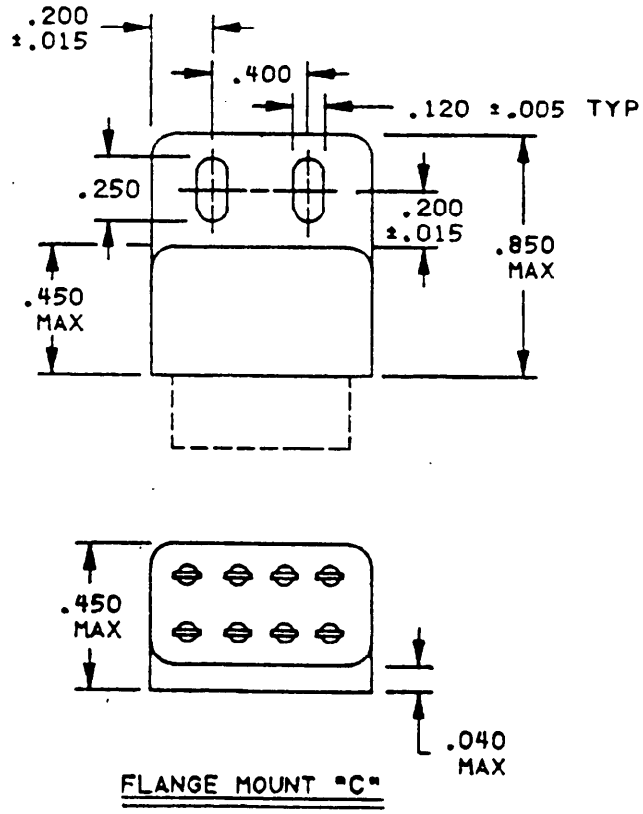
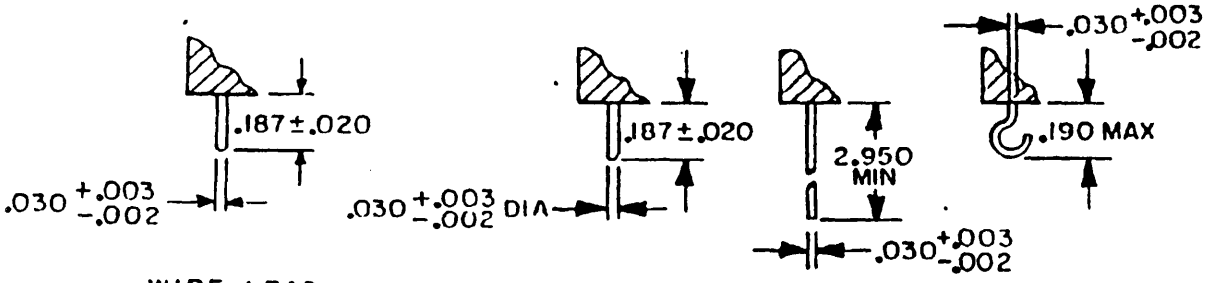


FIGURE 1. Dimensions and configurations - Continued.

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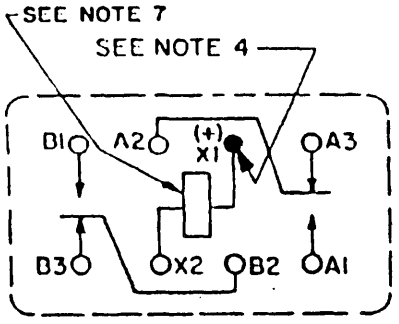


WIRE LEAD
PLUG-IN (PI)
SEE NOTE 9

WIRE LEAD
SOLDER PIN-(SP)

WIRE LEAD

SOLDER LUG



CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm	Inches	mm
.002	0.05	.200	5.08
.003	0.08	.250	6.35
.005	0.13	.400	10.16
.015	0.38	.410	10.41
.020	0.51	.450	11.43
.030	0.76	.600	15.24
.040	1.02	.810	20.57
.100	2.54	.850	21.59
.120	3.05	1.062	26.97
.125	3.18	1.078	27.38
.187	4.75	1.320	33.53
.190	4.83	2.950	74.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Indicated terminal shall be marked with contrasting bead.
5. Circuit diagram (including terminal designations) marked on top if legible from the mounted position, otherwise marking surface is optional.
6. Mounting screw head clearances based on use of .112 (2.84 mm) diameter fillister head screws.
7. Coil symbol shall be in accordance with MIL-STD-1285.
8. Mounting surface finish shall be compatible with aluminum (duralumin type) as defined by the compatible couples table of MIL-R-39016.
9. Finish gold plate in accordance with MIL-G-45204, type II, class I, grade C, except knoop hardness shall be 130 to 240, underplating nickel 50 to 150 microinches thick.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Cleaning and small particle inspection: Applicable.

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc, 0.3 ampere at 115 V ac, (60 and 400 Hz). 1/

Inductive: 0.75 ampere at 28 V dc with 0.200 henry inductance.

Lamp: 0.160 ampere at 28 V dc.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance and voltage drop:

High level:

Before life: 0.050 ohm maximum.

During life: Not more than 5 percent of open circuit voltage.

After life: 0.100 ohm maximum.

Low level:

Before life: 0.050 ohm maximum.

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

Before: 0.050 ohm maximum.

During: 1 ohm maximum.

After: 0.100 ohm maximum.

Contact bounce: 2 milliseconds (ms) maximum (applicable for after high level rated load life and intermediate current tests).

Contact stabilization time: 2.0 milliseconds maximum.

Overload (high level only): Two times rated current.

COIL DATA: See table I.

Operate time: 4 milliseconds maximum over temperature range with rated coil voltage.

Release time: 4 milliseconds maximum over temperature range from rated coil voltage.

1/ 400 Hz load testing is required for qualification only.

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ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	1,000	350 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	1,000	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	1,000	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD 202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable for qualification, group C, and group A inspection testing.

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 G). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Particle impact noise detection: Applicable.

PHYSICAL DATA:

Terminals: See figure 1 and table I.

Terminal strength: 3 ±0.3 pounds pull.

Terminal solderability: Applicable.

Terminal twist test: Applicable to wire lead terminals.

Dimensions and configurations: See figure 1.

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TABLE I. Dash number and applicable characteristics. 1/

Dash number 2/				Mount	Coil voltage 4/ (V dc)		At +25°C				Over temperature 3/ range		
Wire lead (PI) 5/	Solder lug	Wire lead (SP)	Wire lead		Rated	Max	Coil resistance ohms ±10%	Specified pickup voltage (V dc) (max) 3/	Specified hold voltage (V dc) 3/	Specified dropout voltage (V dc) 3/	Specified pickup voltage (V dc) (max)	Specified hold voltage (V dc)	Specified dropout voltage (V dc)
001	002	003	004	A									
---	005	006	007	B									
008	009	010	011	None	26.5	32.0	700	13.5	8.0	1.5	18.0	14.0	1.0
	012			C									
013	014	015	016	A									
---	017	018	019	B									
020	021	022	023	None	12.0	15.0	160	6.4	4.0	0.7	9.0	5.8	0.50
	024			C									
025	026	027	028	A									
---	029	030	031	B									
032	033	034	035	None	6.0	7.5	40	3.2	2.0	0.35	4.5	2.9	0.25
	036			C									
037	038	039	040	A									
---	041	042	043	B									
044	045	046	047	None	5.0	6.0	27	2.7	1.65	0.29	3.8	2.4	0.21
	048			C									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum, or peak ac open circuit, are not recommended for subsequent use in low level applications.

2/ The suffix letter M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations): M, 1.0; P, 0.1; R, 0.01. Example, 004M - - - - 004R.

3/ Pick-up, hold and dropout voltages as shown are for test purposes only and are not to be used for design criterias.

4/ Caution: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

5/ Solderability for plug-in relays is not applicable.

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Weight: 13 grams (0.46 ounce) maximum.

Identification marking (full): Applicable.

Seal: Hermetic, adjunct sealant shall not be used.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.

Low level: 100,000 cycles plus 900,000 mechanical life.

Part or Identifying Number (PIN): M39016/606- (dash number from table I and suffix letter designating failure rate level).

QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Failure rate qualification: The manufacturer shall qualify his products to failure rate level "M" initially.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group submission	
33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.	M39016/606-002	33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.
	M39016/606-019 M39016/606-012	2 units, qualification inspection table, group I, group II, shock, vibration, acceleration, terminal strength, and seal.
	M39016/606-001 M39016/606-008	2 units, qualification inspection table, group I, group II, shock, and vibration.
	M39016/606-034 M39016/606-038	2 units, qualification inspection table, group I, group II.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/6 may be used in addition to MIL-R-39016/606 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

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Qualification inspection (reduced testing) and sample size: If the relays produced for MIL-R-39016/606 are similar in construction and design, except for the additional inspections specified herein, to the relays produced for MIL-R-39016/6, then reduced testing may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/6 at the "M" failure rate level. See table III.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage, group A inspection table. 1 unsealed sample unit for internal examination.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS
Air Force - 99
DLA - ES

User activities:
Navy - MC, OS, SH
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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